

Communicating with Data – Visual Encoding

Dr. Ab Mosca (they/them)

Slides based off slides courtesy of Jordan Crouser (<https://jcrouser.github.io/>)

Event

Interest Meeting Tech Studio

All majors welcome!

Wed Sept 16th

7:30-8:30pm

BASS HALL 204

JOIN A STUDENT-LED TECH
COMMUNITY AT SMITH FOR
PROJECTS, WORKSHOPS, CAREER
EVENTS, AND MORE.

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GOOGLE DEVELOPER GROUPS

Plan for Today

- Mental models: how we process information
- Visualization building blocks
- Mapping data to visuals
- Practice

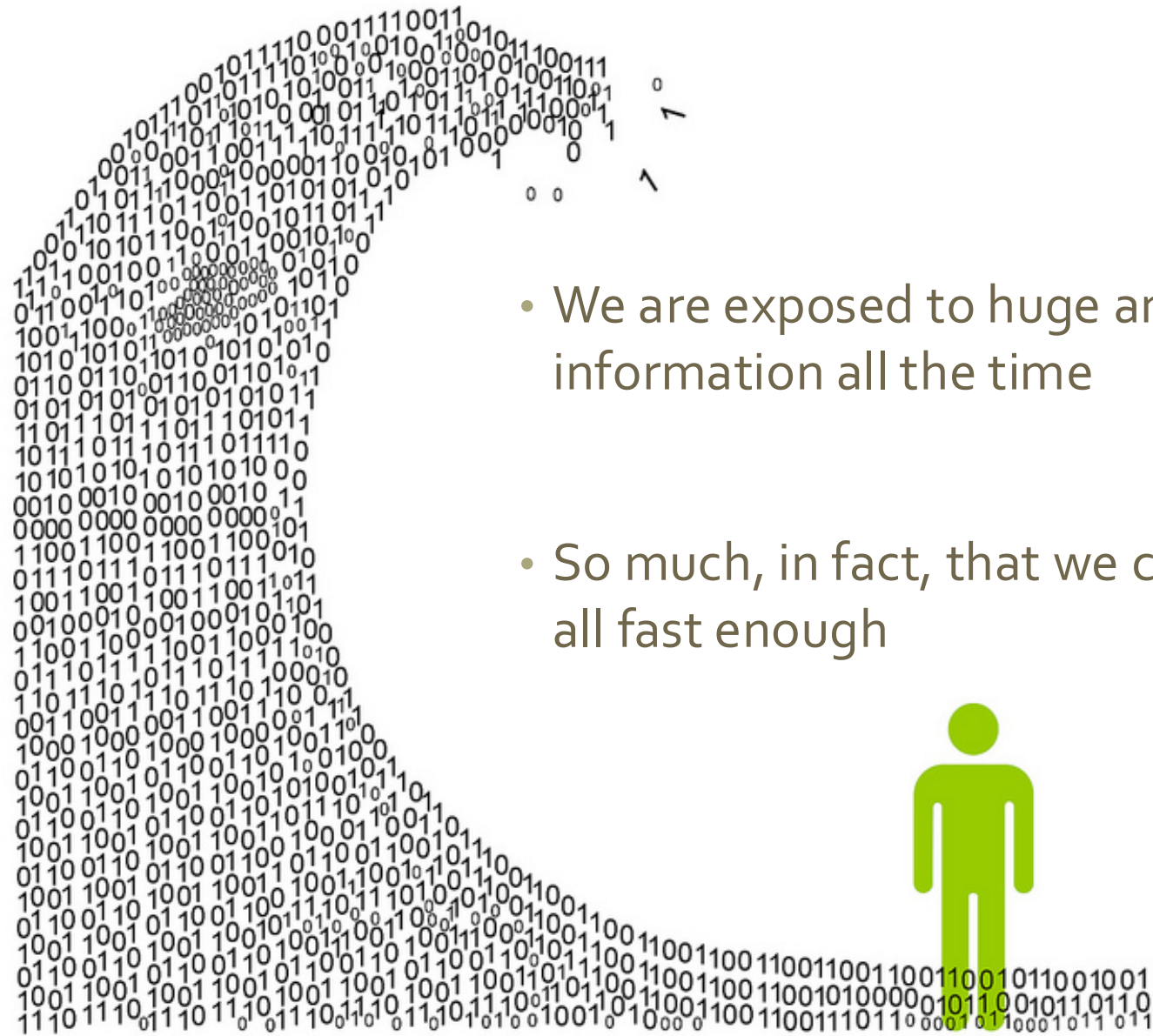
Checking in

- Are you on Slack?
 - All communication will be through Slack (see syllabus for details)
- Are you on Gradescope?
- Have you reviewed the syllabus? (Good way to self-check– do you know the GenAI policy for this class?)

Looking forward

- a1 Due Wednesday! (I will not always remind you of assignment due dates; check the class schedule)
- Find instructions on the course website under the “Assignments” tab
 - Questions? Come to office hours: T 10:30-12:00, 13:30-15:30, and by appointment

Information overload



- We are exposed to huge amounts of information all the time
- So much, in fact, that we can't process it all fast enough

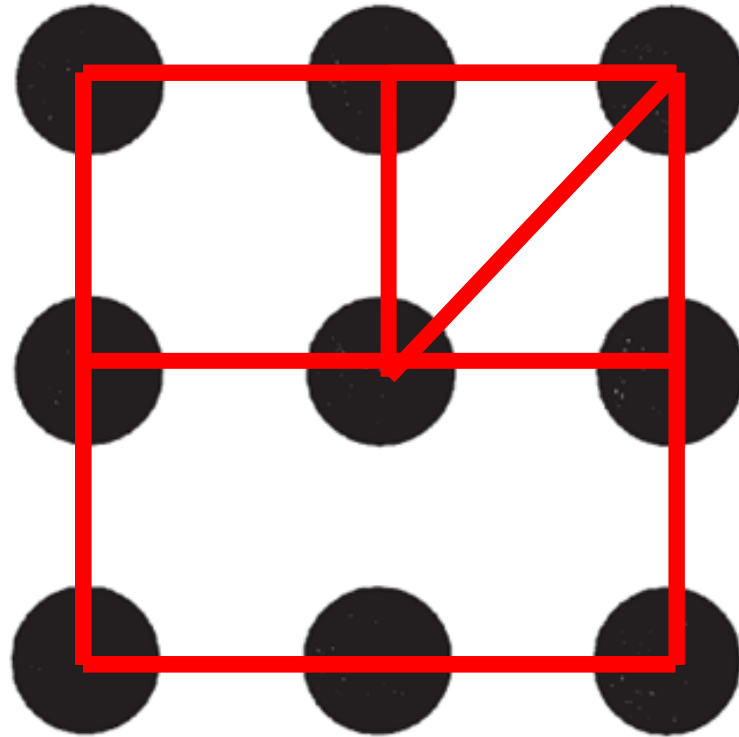
Mental models



To cope, we construct **mental models**:
abstracted, simplified versions of the world
that are more manageable

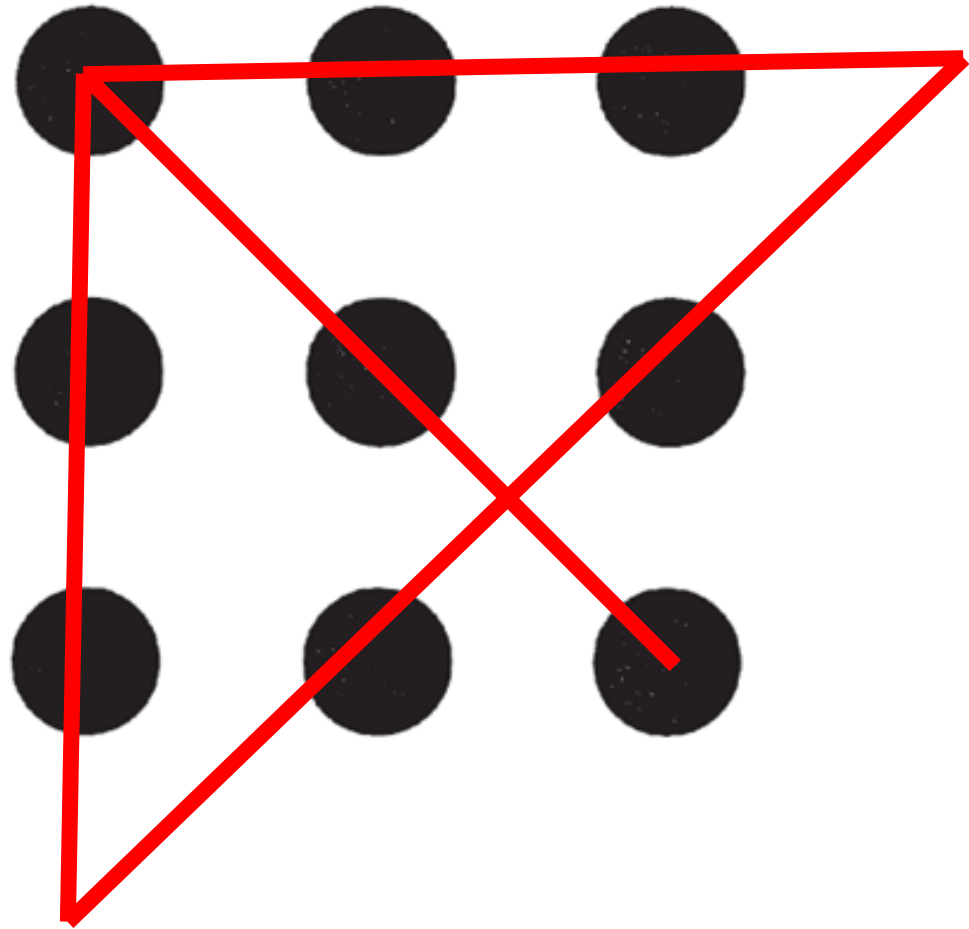
The 9-dot problem

Task 1: Connect all 9 dots using only straight lines



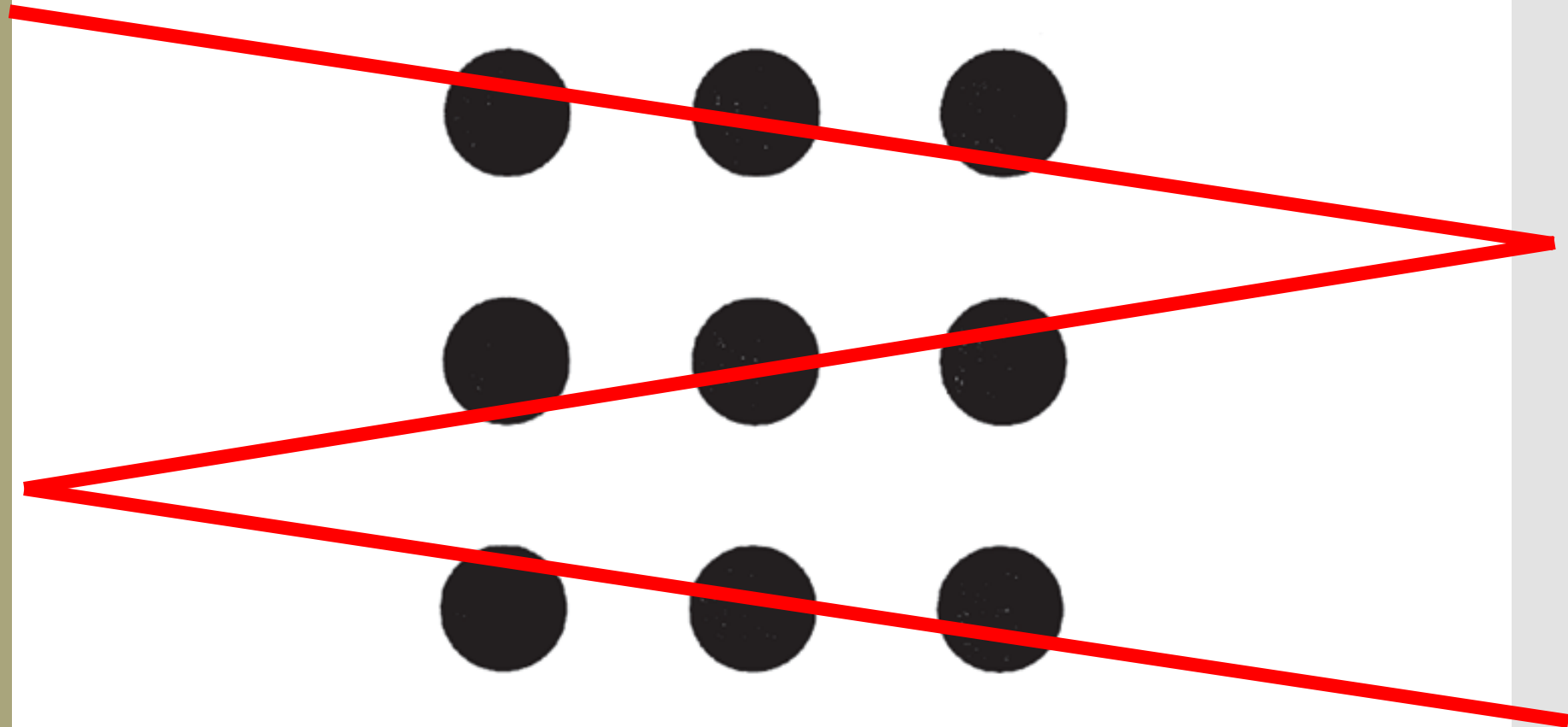
The 9-dot problem

Task 2: Connect all 9 dots using 4 straight lines



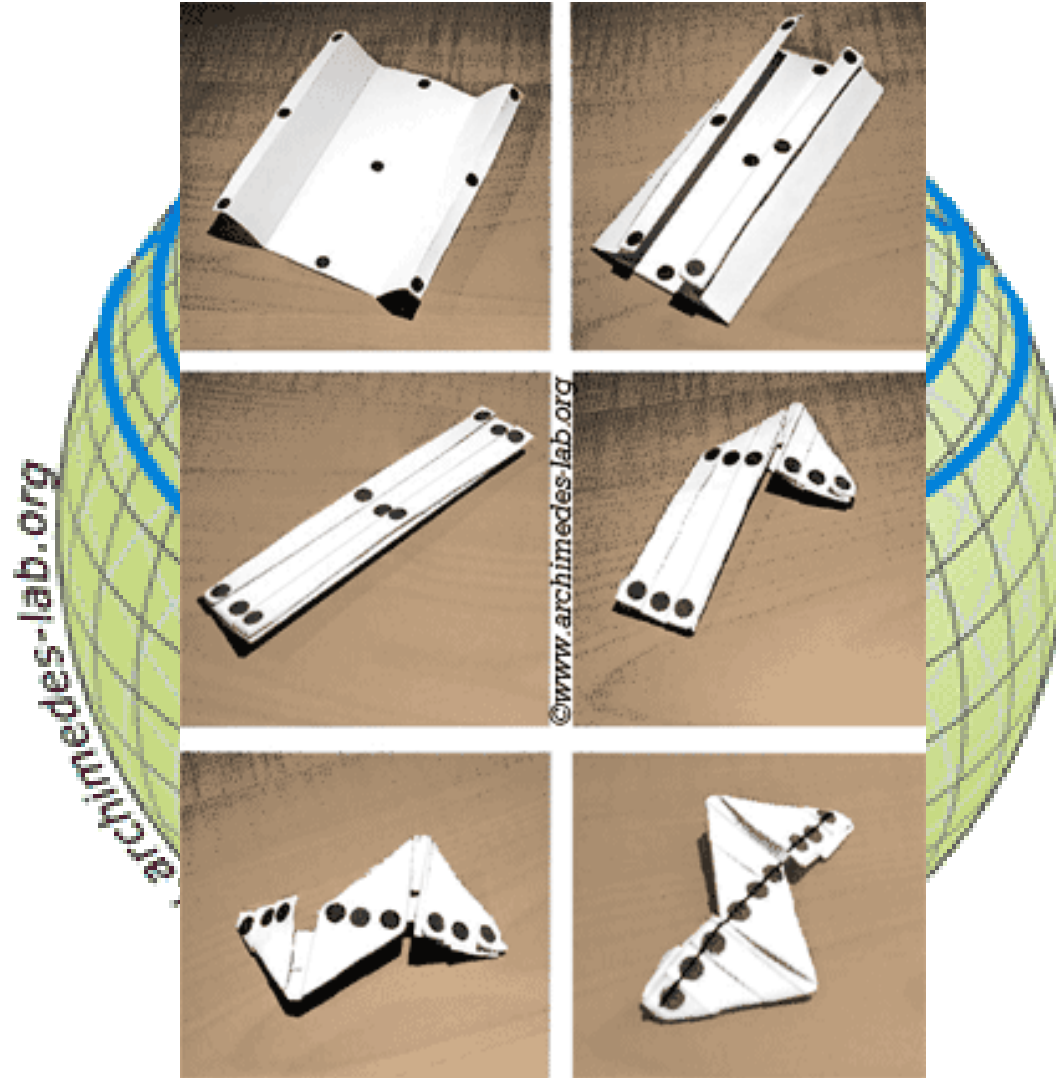
The 9-dot problem

Task 3: Connect all 9 dots using 3 straight lines



The 9-dot problem

Task 4: Connect all 9 dots using 1 straight line



Mental Models: a Sketch



Mental Models

1. We tend to see what we expect to see

- Mental models are built from prior experience
- We expect new input to “fit” the existing model
- Updates are **expensive**: given input that almost fits, we’ll distort information to avoid re-fitting the model
- **Expectation** is at least as strong as perception



Mental Models

2. Mental models form quickly, & update slowly

- “First impressions matter”
- Early information can have the highest impact
- The order in which we present pieces of information can shape how a person understands the whole
- Once a mental model is formed, it takes effort to alter it



Mental Models

3. New information gets incorporated into the existing model

- Integrating competing perspectives is challenging
- Switching between multiple perspectives is also difficult (visually or mentally)
- **Tricky part:** real-world problem often require such perspective switching



Mental Models

4. Initial exposure interferes with accurate perception

- Longer exposure to ambiguous data makes people **more confident** in their initial model
- This is true even if new data presents strong evidence that their model is **wrong!**
- Important: need to be intentional when we design, because incremental information can be **misleading**



Mental Models

The good:

- Well-tuned mental models let us process information quickly
- Frees up more processing power to synthesize information

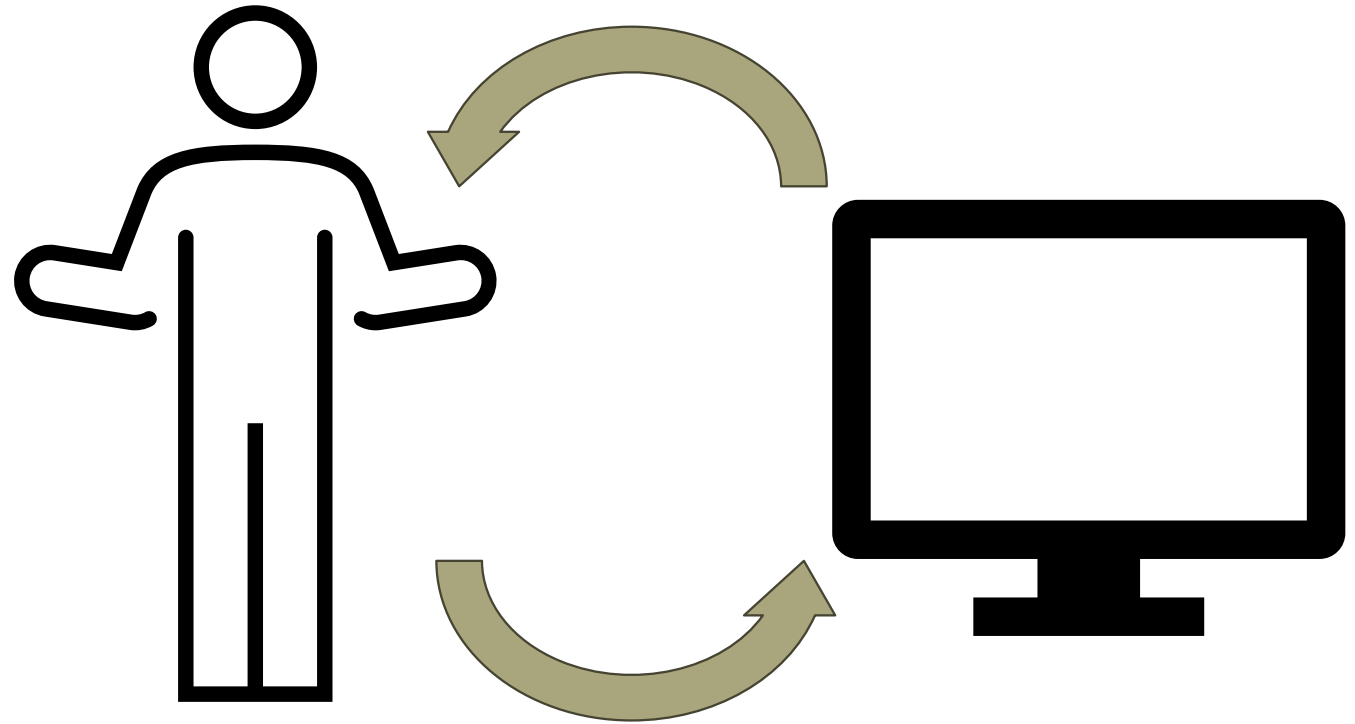
The bad:

- People (esp. experts) tend not to notice information that contradicts their mental model
- A “fresh pair of eyes” can be beneficial

The ugly:

- Mental models are unavoidable: everyone has them, and they're all different
- **Key:** be aware of how mental models form, how they shape perception, and how to support (or challenge) them

So what do we have to work with?



Data

- Remember...

country	year	cases	population
Afghanistan	1999	37745	19987071
Afghanistan	2000	4666	20095360
Brazil	1999	31737	17206362
Brazil	2000	80488	17404898
China	1999	212258	1272915272
China	2000	216766	128042583

variables

country	year	cases	population
Afghanistan	1999	37745	19987071
Afghanistan	2000	4666	20095360
Brazil	1999	31737	17206362
Brazil	2000	80488	17404898
China	1999	212258	1272915272
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observations

country	year	cases	population
Afghanistan	1999	37745	19987071
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values



Data → Visuals

- Remember...

country	year	cases	population
Afghanistan	1999	216745	19987071
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Brazil	1999	31737	17206362
Brazil	2000	80488	17404898
China	1999	212258	127215272
China	2000	216766	128028583

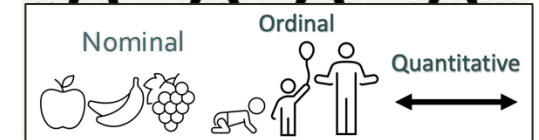
variables

country	year	cases	population
Afghanistan	1999	216745	19987071
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values



- **Big idea behind visualization**

- Data have dimensions
- Visualizations have dimensions, too
- To build good visualizations, we need to **map data dimensions to visual dimensions** in a principled way

Data → Visuals

- Remember...

country	year	cases	population
Afghanistan	1999	216745	19987071
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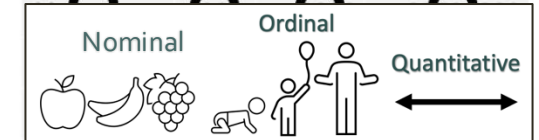
variables

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values



- Big idea behind visualization

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map data dimensions to visual dimensions in a principled way

Data → Visuals

Data

country	year	cases	population
Afghanistan	1999	1866	19987071
Afghanistan	2000	1866	20095360
Brazil	1999	30737	17206362
Brazil	2000	80488	17404898
China	1999	210258	1272015272
China	2000	210766	128608583

variables

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Afghanistan	1999	1866	19987071
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China	1999	210258	1272015272
China	2000	210766	128608583

observations

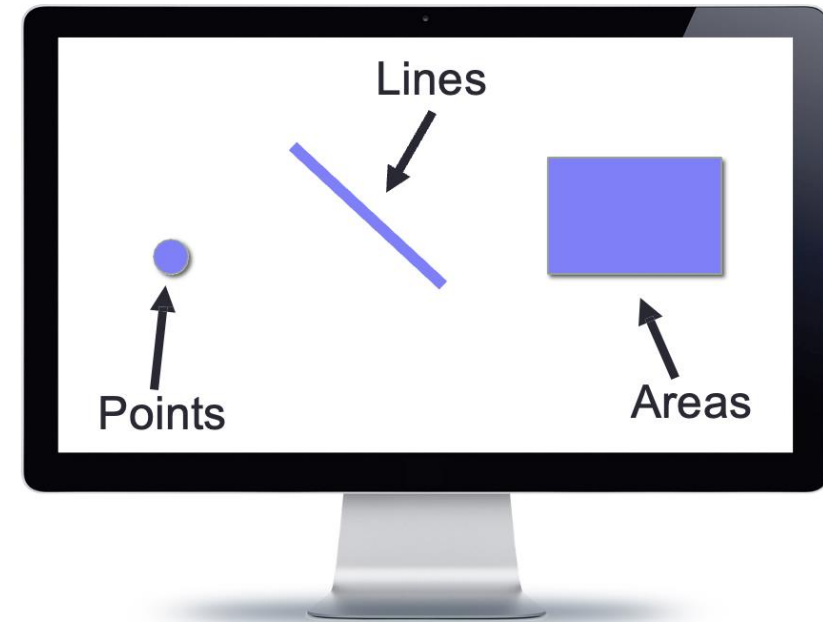
country	year	cases	population
Afghanistan	1999	1866	19987071
Afghanistan	2000	1866	20095360
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values



Visuals

- **Marks**
- The “ink”



Data → Visuals

Data

country	year	cases	population
Afghanistan	1999	1866	19987071
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observations

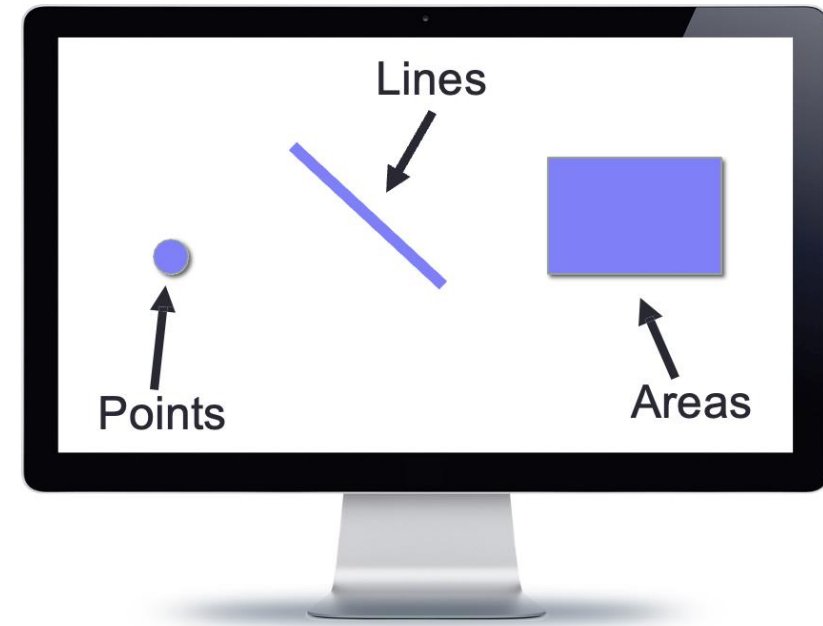
country	year	cases	population
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Brazil	1999	30737	17206362
Brazil	2000	80488	17404898
China	1999	210258	1272015272
China	2000	210766	128608583

values

Nominal: Apple, Banana, Grapes
Ordinal: Person sitting, Person standing, Person jumping
Quantitative: Double-headed arrow

Visuals

- **Marks**
 - The “ink”

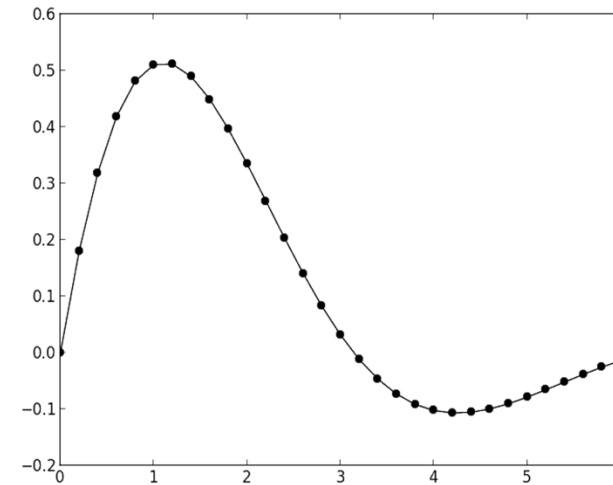
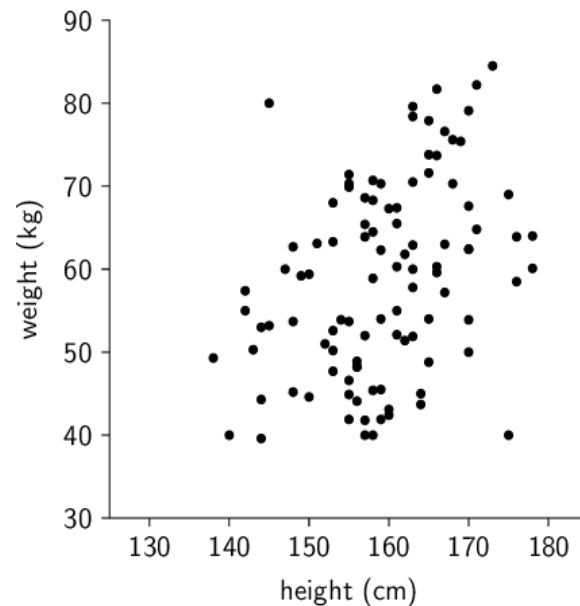


- **Channels or dimensions**
 - *How the marks show up on the page*

Visual Channels / Dimensions

Position

- Encode information using *where* mark is drawn
- Ex.



along a common scale



spatial region

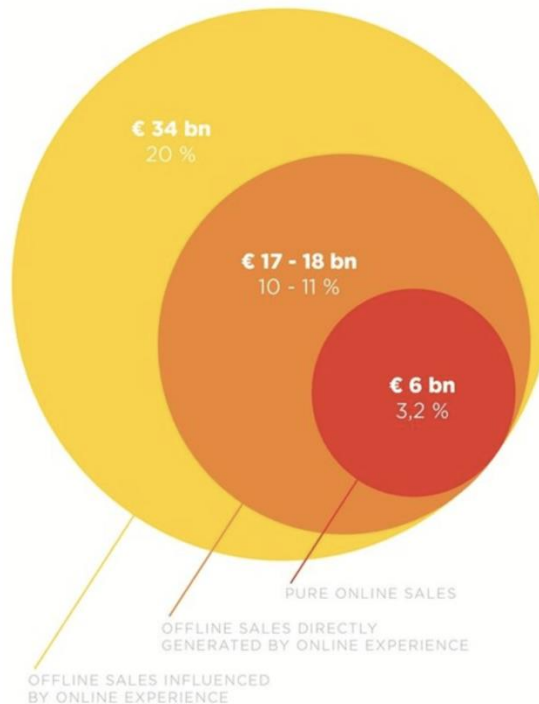
Visual Channels / Dimensions

Size / Area

- Encode information using *how big* mark is drawn

- Ex. **HOW DIGITAL MARKETING INFLUENCES GLOBAL LUXURY SALES**

2011 €bn and
% of market



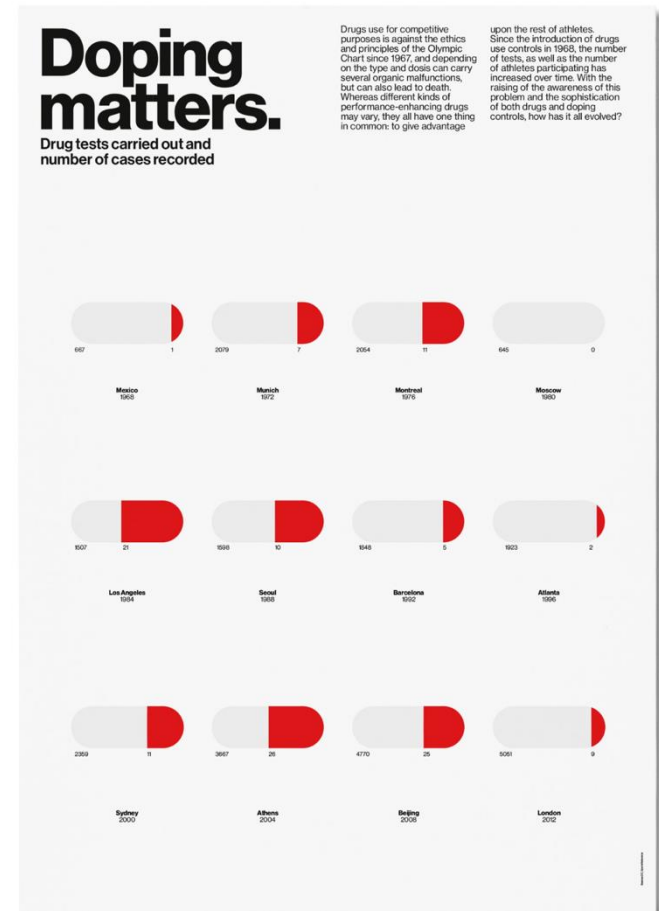
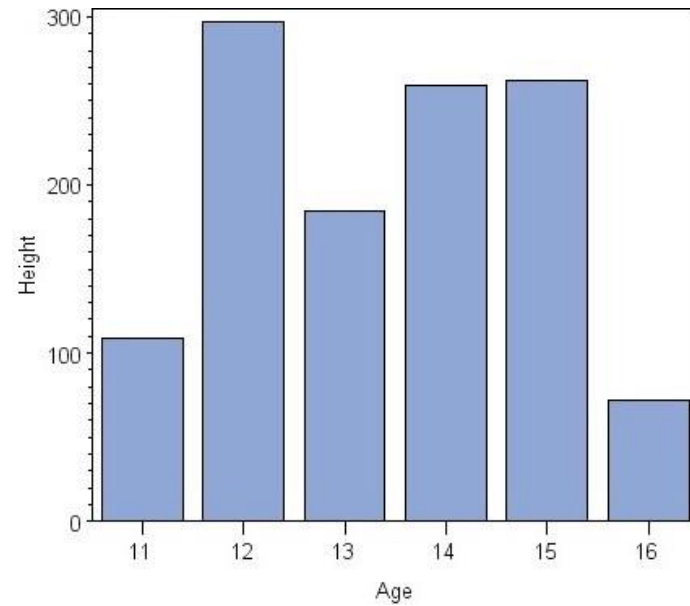
"The Force Awakens" could produce **\$9.6 billion** in revenue from worldwide ticket sales, merchandise, and home entertainment in roughly the first year of release.

\$5.0B Merchandise	\$500M Disney's Infinity 3.0 Videogame	\$214M TV licensing (intl.)
	\$780M Star Wars Battlefront Videogame	\$235M TV licensing (domestic)
		\$458M DVDs/downloads
	\$1.65B Box office sales (international)	

Visual Channels / Dimensions

Length

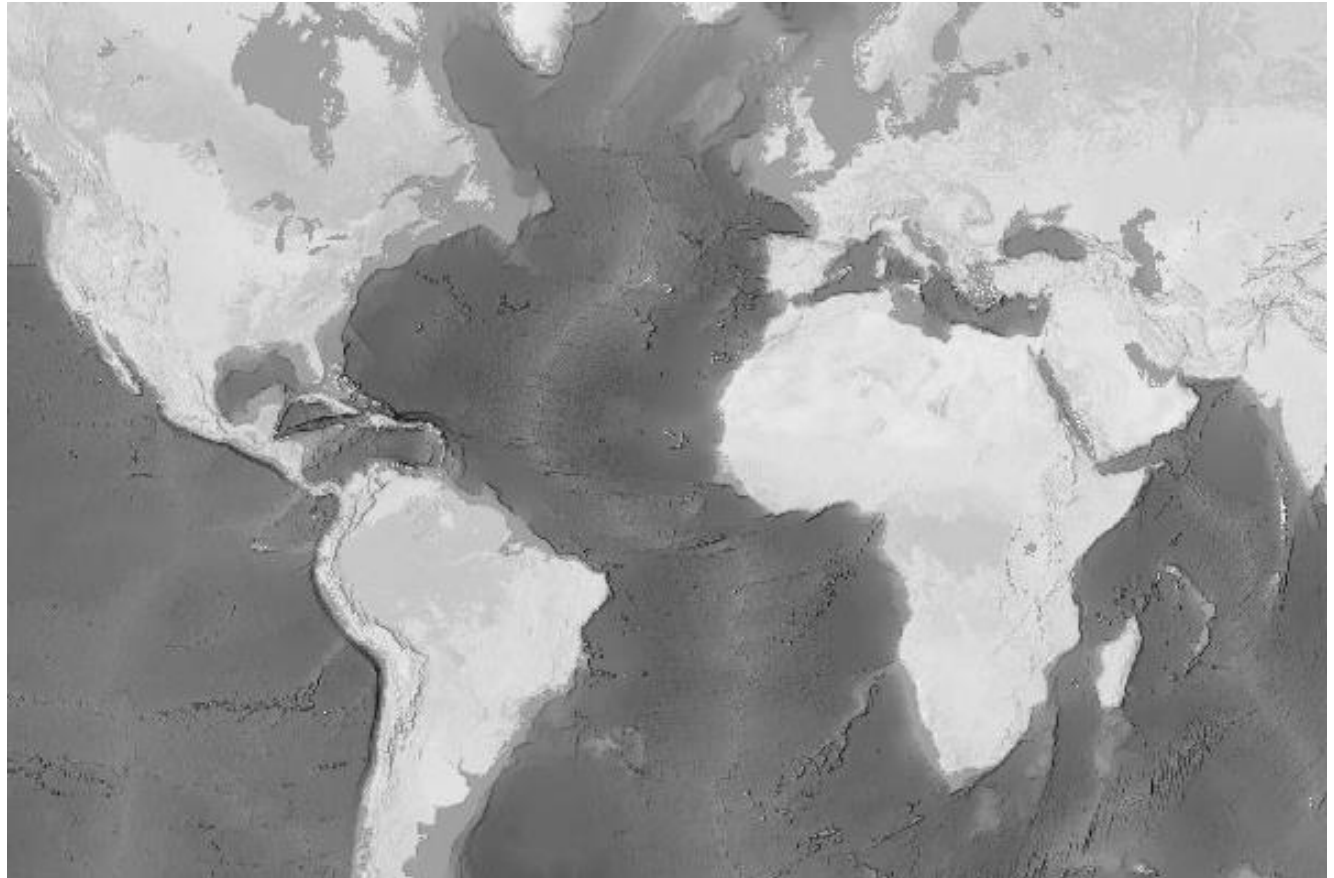
- Encode information using *how long* mark is drawn
- Ex.



Visual Channels / Dimensions

Color: Luminance

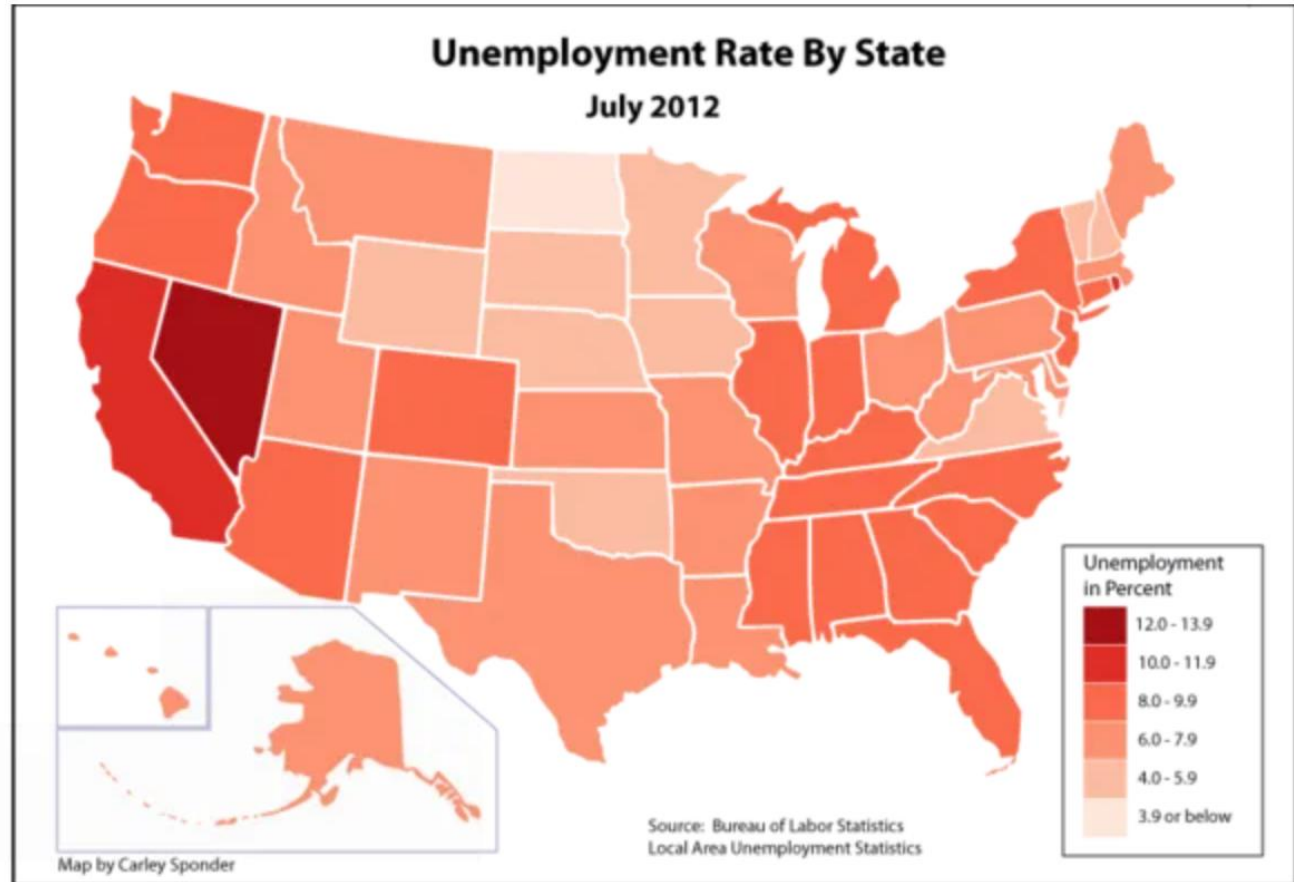
- Encode information using *how dark* mark is drawn
- Ex.



Visual Channels / Dimensions

Color: Saturation

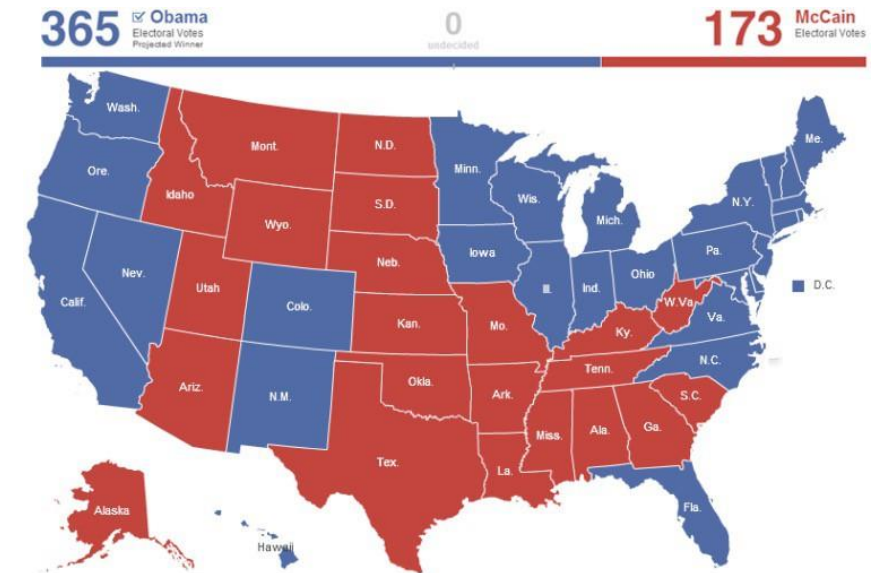
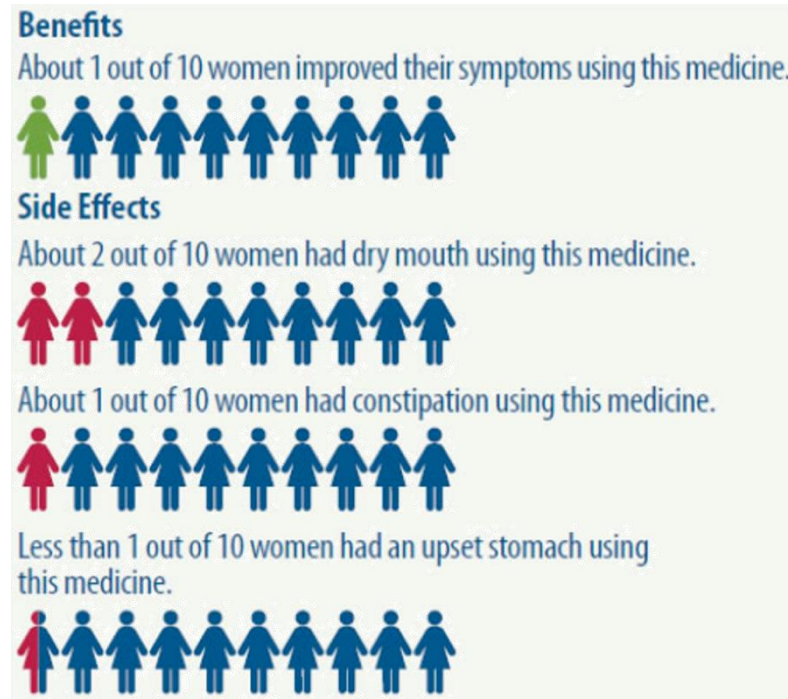
- Encode information using *how much color* mark has
- Ex.



Visual Channels / Dimensions

Color: Hue

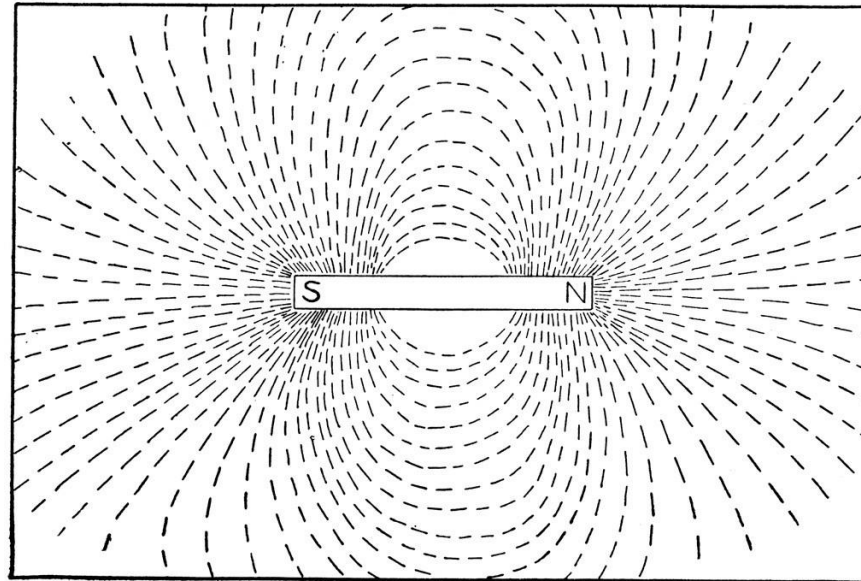
- Encode information using *hue* of mark
- Ex.



Visual Channels / Dimensions

Orientation / Tilt / Angle

- Encode information using how mark is *rotated*
- Ex.



Visual Channels / Dimensions

Shape

- Encode information using how mark is *shaped*
- Ex.

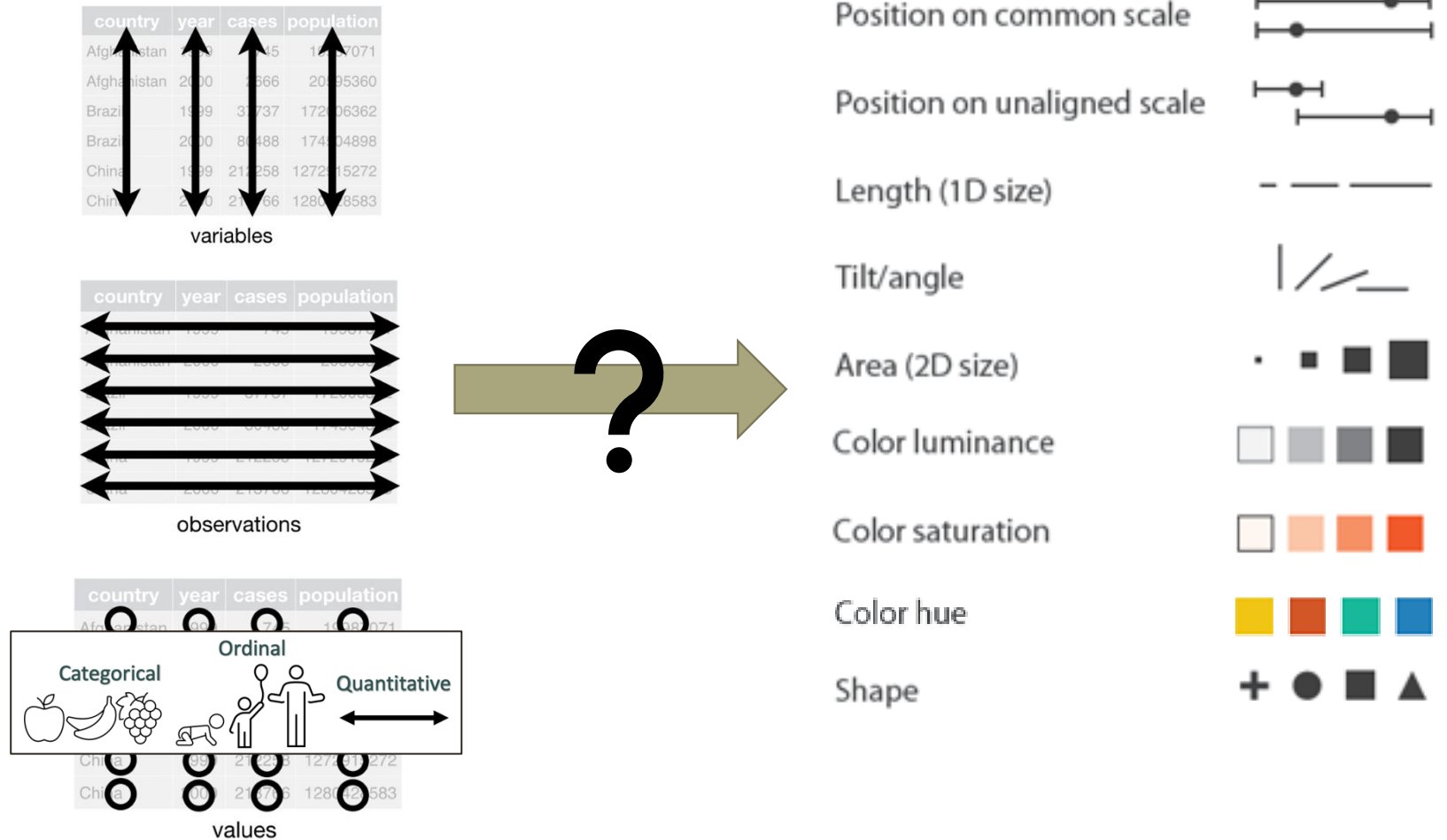


Data → Visuals

- Remember... **Big idea behind visualization**
 - **Visual Encoding:** Map data dimensions to visual dimensions in a principled way

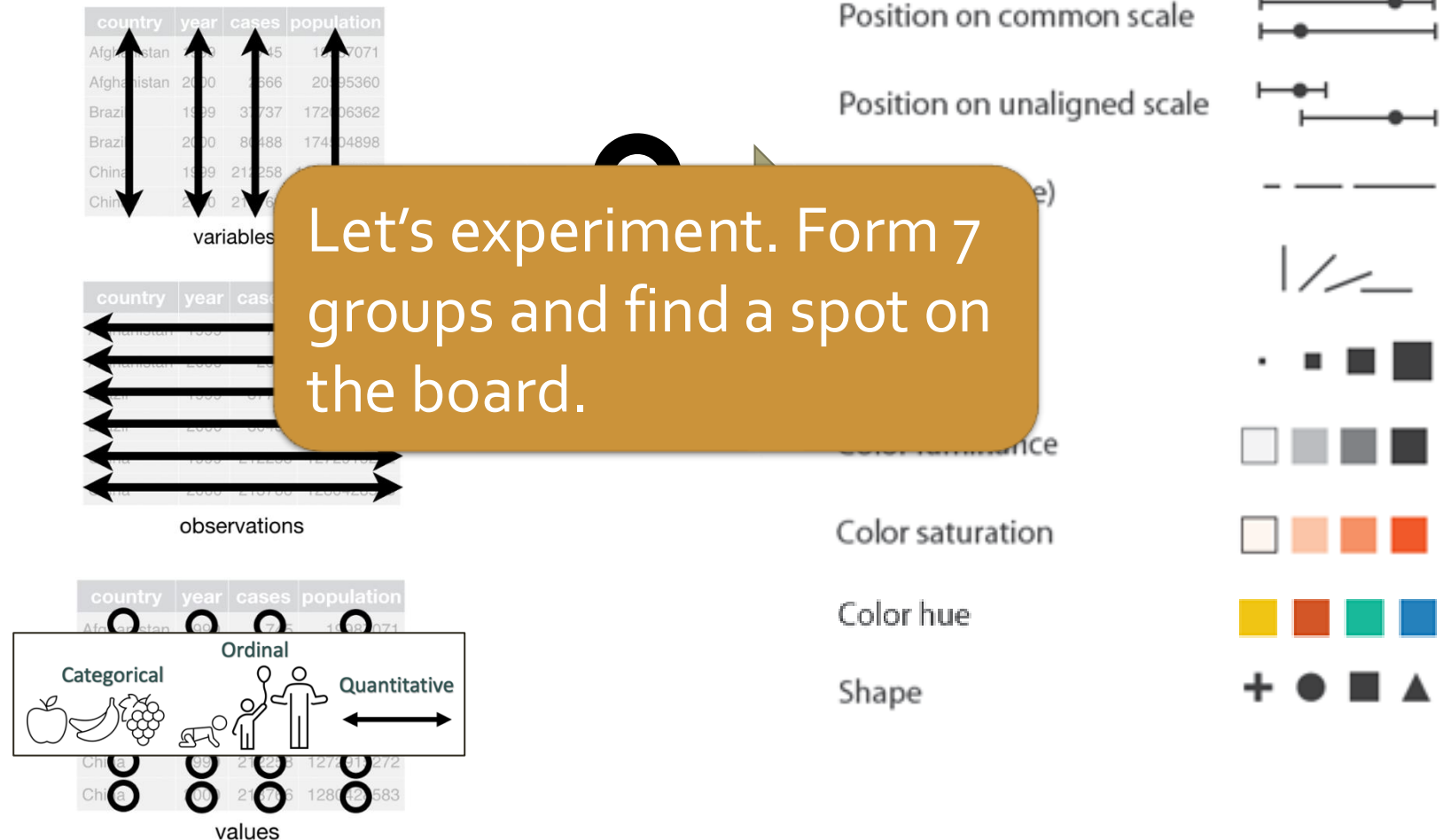
Data → Visuals

- Remember... **Big idea behind visualization**
 - Visual Encoding:** Map data dimensions to visual dimensions in a principled way



Data → Visuals

- Remember... **Big idea behind visualization**
 - Visual Encoding:** Map data dimensions to visual dimensions in a principled way



Let's experiment. Form 7 groups and find a spot on the board.

Work with your group to represent each observation in this dataset as a point (mark) styled only using the visual channel you were assigned.

Your goal is for other groups to easily infer which point represents which observation.

Data → Visuals

Name
blueberry
asparagus
pumpkin
pea
watermelon
long bean

Position on common scale	
Position on unaligned scale	
Length (1D size)	
Tilt/angle	
Area (2D size)	
Color luminance	
Color saturation	
Color hue	
Shape	

Work with your group to represent each observation in this dataset as a point (mark) styled using the visual channel you were assigned.

Your goal is for other groups to easily infer which point represents which observation.

Data → Visuals

Peak
July
April
October
June
August
September

Position on common scale	
Position on unaligned scale	
Length (1D size)	
Tilt/angle	
Area (2D size)	
Color luminance	
Color saturation	
Color hue	
Shape	

Work with your group to represent each observation in this dataset as a point (mark) styled using the visual channel you were assigned.

Your goal is for other groups to easily infer which point represents which observation.

Data → Visuals

Price (per lb)
6.45
4.99
0.24
7.00
3.65
3.64

Position on common scale	
Position on unaligned scale	
Length (1D size)	
Tilt/angle	
Area (2D size)	
Color luminance	
Color saturation	
Color hue	
Shape	

Data → Visuals

What type of variable is Name?

Name
blueberry
asparagus
pumpkin
pea
watermelon
long bean

Position on common scale	
Position on unaligned scale	
Length (1D size)	
Tilt/angle	
Area (2D size)	
Color luminance	
Color saturation	
Color hue	
Shape	

Data → Visuals

What type of variable is Peak?

Peak
July
April
October
June
August
September

Position on common scale	
Position on unaligned scale	
Length (1D size)	
Tilt/angle	
Area (2D size)	
Color luminance	
Color saturation	
Color hue	
Shape	

Data → Visuals

What type of variable is Price?

Price (per lb)
6.45
4.99
0.24
7.00
3.65
3.64

Position on common scale	
Position on unaligned scale	
Length (1D size)	
Tilt/angle	
Area (2D size)	
Color luminance	
Color saturation	
Color hue	
Shape	

Data → Visuals

- Remember... **Big idea behind visualization**
 - Visual Encoding:** Map data dimensions to visual dimensions in a principled way
 - Insight 1:** Not all visual dimensions can represent all data types

Position on common scale	
Position on unaligned scale	
Length (1D size)	
Tilt/angle	
Area (2D size)	
Color luminance	
Color saturation	
Color hue	
Shape	

Nominal 	Ordinal 	Quantitative 
✓	✓	✓
✓	✓	✓
	✓	✓
	✓	✓
	✓	✓
	✓	✓
	✓	✓
✓		
✓		

Data → Visuals

- Remember... **Big idea behind visualization**
 - **Visual Encoding:** Map data dimensions to visual dimensions in a principled way
 - **Insight 1:** Not all visual dimensions can represent all data types

➔ **Magnitude** Channels: **Ordered** Attributes

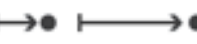
Position on common scale 

Position on unaligned scale 

Length (1D size) 

Tilt/angle 

Area (2D size) 

Depth (3D position) 

Color luminance 

Color saturation 

Curvature 

Volume (3D size) 

➔ **Identity** Channels: **Categorical** Attributes

Spatial region 

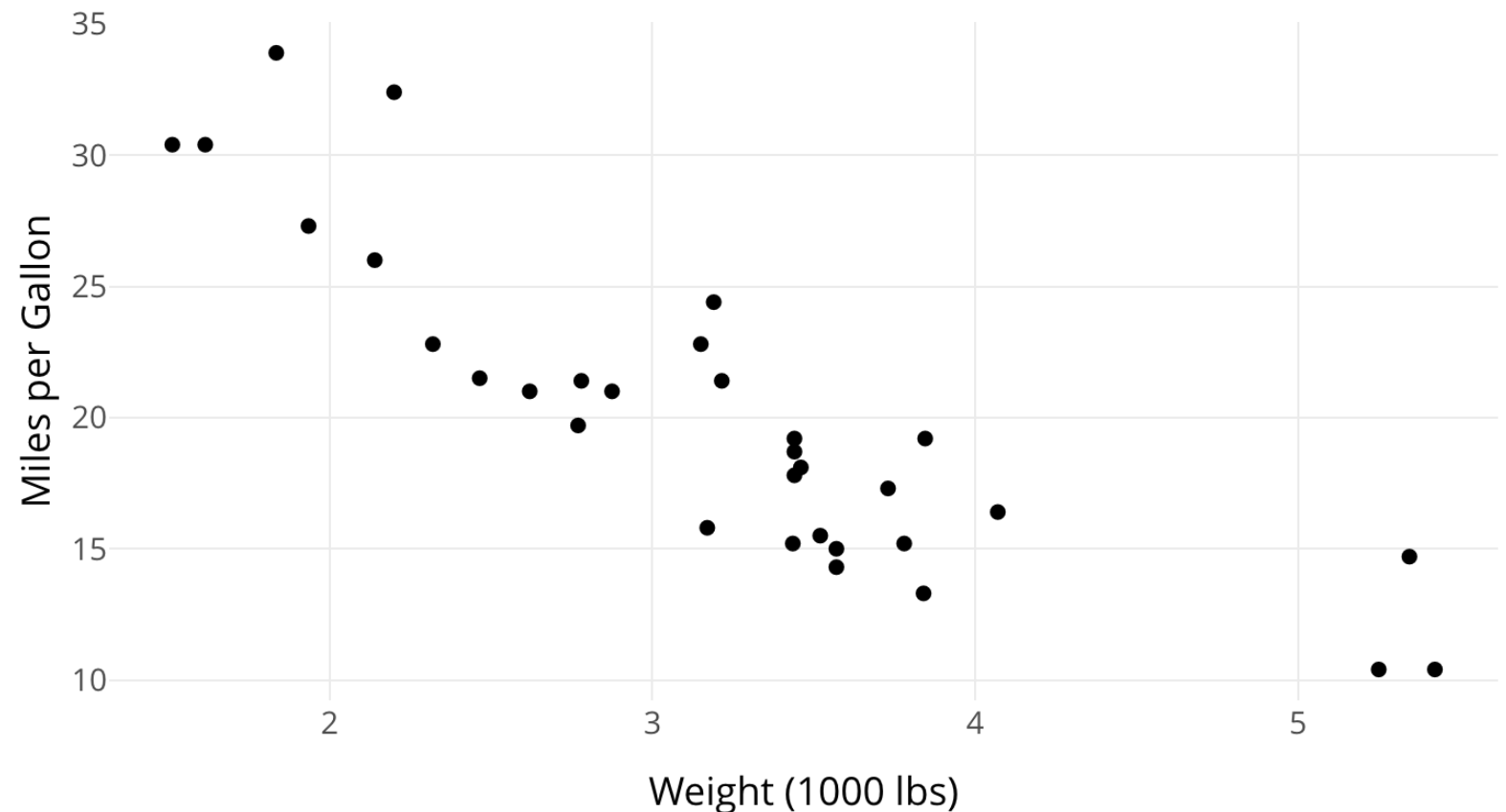
Color hue 

Motion 

Shape 

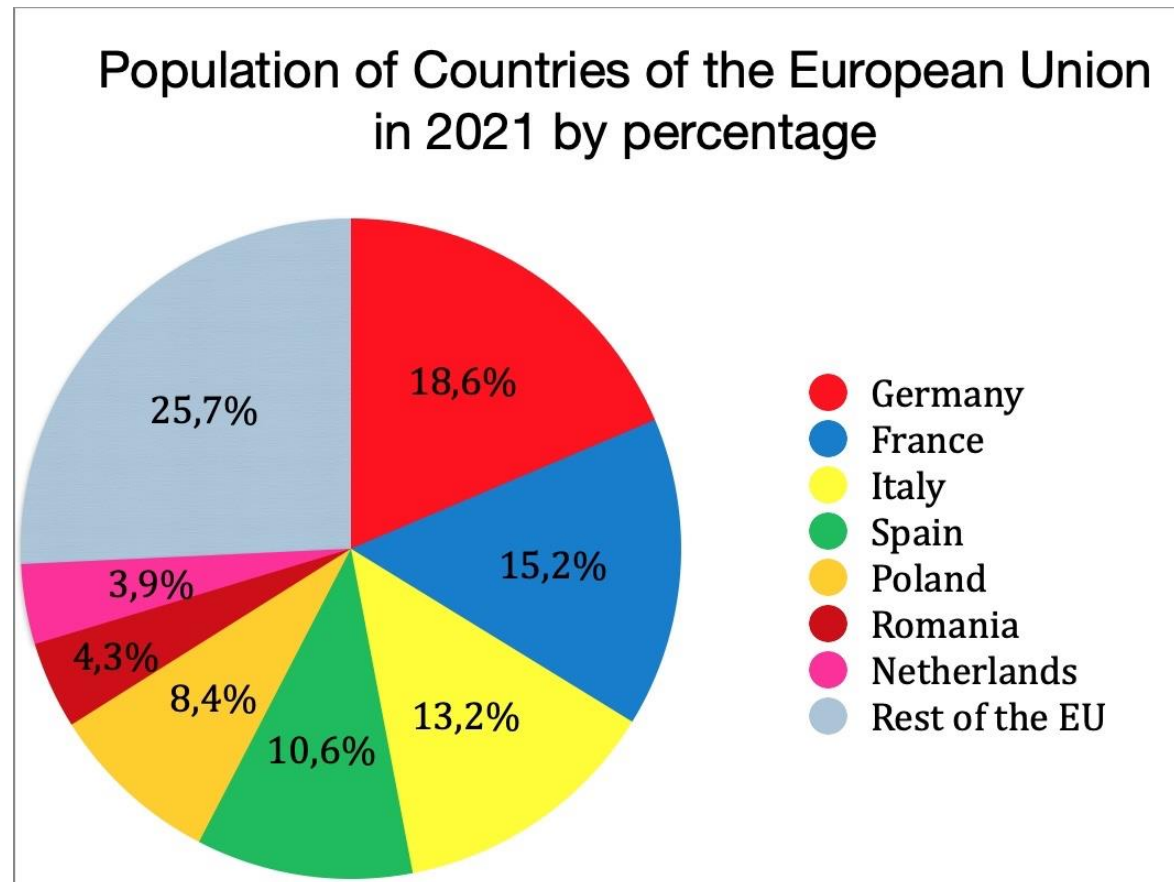
Practice

- What are the marks in this visualization?
 - What variables are shown?
 - What visual channel(s) encode(s) each variable?
- Based on data type and channel type, is the encoding appropriate?



Practice

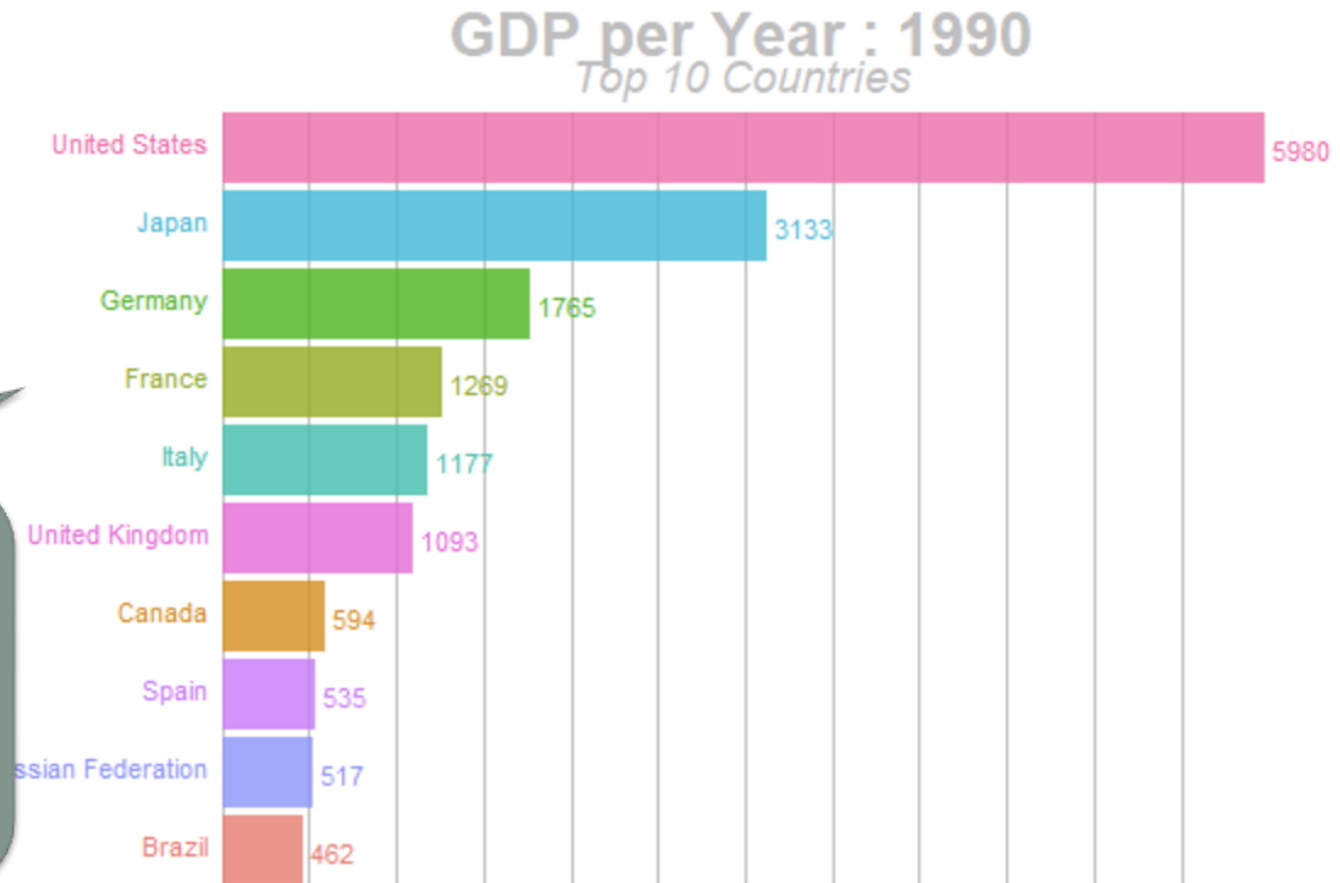
- What are the marks in this visualization?
 - What variables are shown?
- What visual channel(s) encode(s) each variable?
- Based on data type and channel type, is the encoding appropriate?



Practice

Dual (or double) encoding ==
encoding one variable via
more than one channel

- What are the marks in this visualization?
 - What variables are shown?
 - What visual channel(s) encode(s) each variable?
- Based on data type and channel type, is the encoding appropriate?



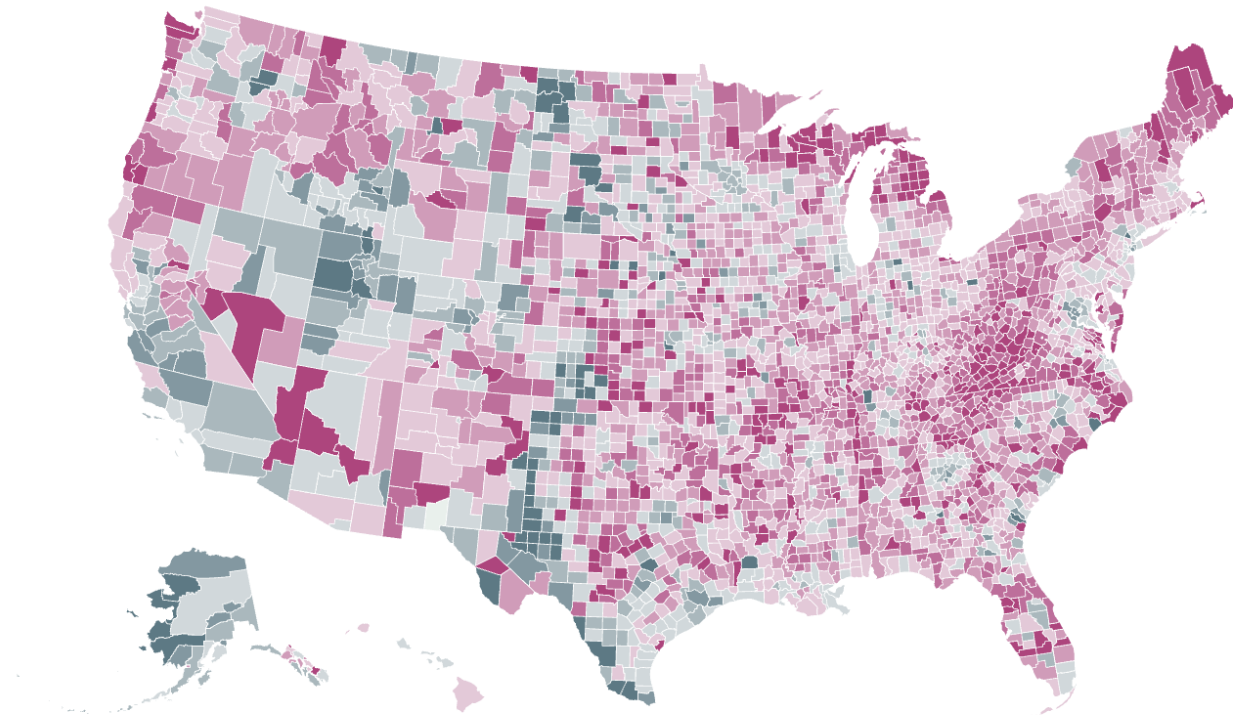
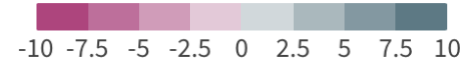
Practice

- What are the marks in this visualization?
 - What variables are shown?
- What visual channel(s) encode(s) each variable?
- Based on data type and channel type, is the encoding appropriate?

In the US natural change rate in population has an East/West divide

Much of California, Texas and Utah are experiencing natural growth while in the east the opposite is true.

Birth rate-Death rate 2021



Source: [U.S. Census Bureau 2021 boundaries](#), [U.S. Census Bureau 2021 population estimates](#)

<https://flourish.studio/examples/?Industry=Featured>

Practice

- How could you visually represent the following data?
Sketch your idea

Total Time (minutes)
51
30
54
30
30
30
80
30
60
40
30

Practice

- How could you visually represent the following data?
Sketch your idea

Total Time (minutes)	Number of Squirrels
51	12
30	24
54	16
30	15
30	28
30	34
80	11
30	16
60	25
40	51
30	59

Practice

- How could you visually represent the following data?
Sketch your idea

Total Time (minutes)	Number of Squirrels	Weather
51	12	43 degrees, sunny
30	24	cold, clear
54	16	43 degrees
30	15	43 degrees, clear
30	28	
30	34	42 degrees, clear
80	11	43 degrees, sunny
30	16	40 degrees, sunny
60	25	45 degrees, sunny
40	51	45 degrees, sunny with shade spots
30	59	